COMPATIBILITY OF PROPER PUSHFORWARD WITH COMPOSITION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as three spans, lines 8-12 (the verbatim package together with the project's own reference, slogan and history display-environment definitions, including the author's commented-out older form of the reference environment), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). The lines 8-12 span is required because a counted statement or its surrounding text may use those display environments, which the project typesets as comments. Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- The degree of a dominant morphism between integral schemes of equal dimension, and multiplicativity of degrees. Same chapter chow.tex, label lemma-equal-dimension (proof line 1944, claimed as S14-algebra-equal-dimension); Morphisms, label morphisms-lemma-degree-composition (proof line 1951). The first makes the degree of the induced morphisms on the images available, and the second gives the multiplicativity of degrees used to compare the two products.

Source reference. Original lemma 02R1: sources/stacks-chow/pinned-original-source.tex, source lines 1816–1816. Navigation pointer for \ref{lemma-equal-dimension} used inside the counted span (line 1944); the argument that consumes it is retained in the span.

Source reference. Original lemma 02NZ: sources/stacks-chow/tags.txt, source lines 3498–3498. Navigation pointer for \ref{morphisms-lemma-degree-composition} used inside the counted span (line 1951); the argument that consumes it is retained in the span.

Source reference. Original reference 02QL: sources/stacks-chow/pinned-original-source.tex, source lines 1372–1372. Navigation pointer for \ref{situation-setup} used inside the counted span (line 1925); the argument that consumes it is retained in the span.

Date: stacks-project:chow.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let (S, δ) be as in Situation 02QL. Let X, Y, and Z be locally of finite type over S. Let $f : X \to Y$ and $g : Y \to Z$ be proper morphisms. Then $g_* \circ f_* = (g \circ f)_*$ as maps $Z_k(X) \to Z_k(Z)$.*

Proof. Let $W \subset X$ be an integral closed subscheme of dimension k. Consider $W' = f(W) \subset Y$ and $W'' = g(f(W)) \subset Z$. Since f, g are proper we see that W' (resp. W'') is an integral closed subscheme of Y (resp. Z). We have to show that $g_*(f_*[W]) = (g \circ f)_*[W]$. If $\dim_\delta(W'') < k$, then both sides are zero. If $\dim_\delta(W'') = k$, then we see the induced morphisms

$$W \longrightarrow W' \longrightarrow W''$$

both satisfy the hypotheses of Lemma 02R1. Hence

$$g_*(f_*[W]) = \deg(W/W') \deg(W'/W'')[W''], \quad (g \circ f)_*[W] = \deg(W/W'')[W''].$$

Then we can apply Morphisms, Lemma 02NZ to conclude. □